

Project 2: Planning Document

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1 WHAT DO YOU WANT TO ACCOMPLISH?

Again, I would like to build on from what I learned from Project 1 (which is based on an earlier light project) in order to create a realistic looking outdoor scene. The final goal of the project is to make use of Three JS in order to create a scene with a realistic looking sun, sky, ground, and overall outdoor environment. This is, of course, a very ambitious project and so I will want to break this up into three smaller projects.

The first part of the project, project 1, dealt with learning about shaders, learning Joshua Koo's and Simon Wallner's demos, and implementing (and even improving upon!) a shader that combines Mie and Rayleigh scattering in order to create realistic outdoor lighting that takes the sunlight and the skylight into consideration for the light computations.

The second part of the project (this project) will involve me using my knowledge about height maps and procedural terrain generation to build terrain in the sky dome that was created in project 1 and researching bump, normal, and parallax mapping to simulate rocky or "dimpled" terrain. I am aware, however, that some lighting problems may occur due to the sky shader that I am using, and I may have to also experiment with the lighting.

The final part of the project, project 3, will focus on the finishing touches of the scene: the lighting, some firefly particles, and a simple body of water that uses environment mapping along with anything else I can find that makes the water more realistic.

2 STATE A QUESTION THAT YOU WILL USE TO DRIVE YOUR EXPLORATION.

Some good questions to drive my exploration of this project is: "How can I apply a bump, normal, or parallax map to a procedurally created terrain in Three JS? How can I accomplish this using a shader?"

3 HOW WILL YOUR GOAL IN 1 EXTEND YOUR UNDERSTANDING OF COMPUTER GRAPHICS?

The last project exposed me to shaders and helped me understand the differences and similarities between vertex and fragment shaders along with learning some new physical concepts. However, it focused primarily on modifying the fragment shader. This project will extend my knowledge of vertex shaders and, of course, bump, normal, and parallax mapping and will help me cement my understanding of the primary differences between them.

4 WHAT RESOURCES/DEMOS/PAPERS/TEXTBOOKS WILL YOU USE TO AID YOU?

I will use quite a few WebGL demos including (but not limited to): Rob Chadwick's Terrain Editor, Three JS "Ninja Head" demo, WhiteFlashWhiteHit's moon demo, and a Kami Pixel demo. I will also use Eric Haines' Udacity course (specifically the lesson on textures), the Angel OpenGL text (chapter 9 includes an example of normal mapping in a shader), and a couple of academic papers.

Angel, E. (2009). *Interactive Computer Graphics: A Top-Down Approach Using OpenGL*. Pearson Education, Inc.
Chapter 9: Shaders

Chadwick, R. (n.d.). *Terrain Editor*. Retrieved from Rob Chadwick: <http://robchadwick.com/TerrainEditor/>

Haines, E. (n.d.). *Interactive 3D Graphics*. Retrieved from Udacity: <https://www.udacity.com/course/cs291>. Lesson 8: Textures and Reflections

McGuire, M. (n.d.). *Steep Parallax Mapping*. Retrieved from <http://graphics.cs.brown.edu/games/SteepParallax/mcguire-steeparallax.pdf>

Moon WebGL. (n.d.). Retrieved from WhiteFlashWhiteHit:
http://whiteflashwhitehit.com/content/2011/02/moon_webgl.html

Pixel Normals. (n.d.). Retrieved from MattDesl GitHub: <http://mattdesl.github.io/kami-demos/release/normals-pixel.html>

Villar, J. (n.d.). *OpenGL Shading Language Course*. Retrieved from OpenGL:
https://www.opengl.org/sdk/docs/tutorials/TyphoonLabs/Chapter_4.pdf. *Pages 15 - 19*.

Web GL Map Demo. (n.d.). Retrieved from Three JS: http://threejs.org/examples/webgl_materials_normalmap.html

5 HOW WILL YOU KNOW YOU'VE DONE A SATISFACTORY JOB?

I know that I have done a satisfactory job when I have some generated terrain in my sky dome that has some type of displacement mapping (be it normal, bump mapping, or parallax mapping) on its texture that is placed via a shader that I created along with having a better understanding of displacement mapping.